RethinkDB: A NoSQL Database *for* Real-Time Applications

Faster than conventional databases, RethinkDB works in real-time. This open source, distributed and document-oriented database is designed to store JSON documents in an operable format with sharding and replication.

Traditional database management systems have similar structures and share common methods to insert, delete, update and query information. NoSQL based database systems, however, offer developers many choices for their specific data storage requirements. New scalability features have revolutionised these databases, though most NoSQL systems still rely on the creation of a specific structure that is organised collectively into a record of data. The access models of NoSQL based systems have not adapted to modern Web applications when it comes to fetching information, adding records to data and getting the information pulled out; rather, the user just queries the database by specifying some significant values.

RethinkDB takes an entirely novel approach to creating a database structure, as well as in the techniques it uses to store and retrieve information. RethinkDB works in real-time, and is an open source, distributed and document-oriented database designed to store JSON documents in an operable format with sharding and replication. It provides real-time pushing of JSON data to the server, which entirely redefines real-time Web application development. It implements a proprietary, function-based query language called ReQL to interact with its schemaless JSON data collections. Like MongoDB, documents in RethinkDB are hierarchical, dynamically typed and schemaless objects.

RethinkDB makes use of a custom-built storage engine primarily based on the Binary Tree File System (BTFRS) from Oracle, which promises various significant advantages such as less overhead on the CPU, SSD optimisation, power failure recovery, support for MVCC, and efficient multi-core operation and data consistency when failures occur.

RethinkDB architecture

RethinkDB consists of different components like a cluster, query execution engine, file system storage, push changes and RethinkDB client drivers.

- **Client drivers:** RethinkDB provides client drivers for various popular programming languages like Ruby, Python, Java, etc.
- **Query engine:** This database has an advanced query handler to perform all sorts of query executions and give results back to the user. It performs queries on the basis of various operations like indexing, sorting, cluster searching and data merging.
- **Clusters:** As RethinkDB is a distributed database, the entire distribution is managed through clustering—sharding or replication.